Intelligence Hub - How to Run & Where to Store PDFs

1. Setup

```
cd c:\<Your Location>\projects\intelligence-hub


# Install dependencies

pip install -r requirements.txt


# Set your API key in .env (Groq free tier recommended)

# Copy .env.example to .env and add your key
```

The .env.example shows you need a GROQ_API_KEY. The code also supports GEMINI_API_KEY, OPENAI_API_KEY, or ANTHROPIC_API_KEY - it uses whichever it finds.

2. Where to Store PDFs

PDFs go in the sample_docs/ folder. It already has 3 sample PDFs:

- cloud_computing.pdf
- neural_networks.pdf
- sustainable_energy.pdf

You can also pass **any path** to a PDF - it doesn't have to be in sample_docs/.

3. How to Run

```
# Basic run with a sample PDF

python project_intelligence_hub.py sample_docs/cloud_computing.pdf


# With custom budget and evaluation

python project_intelligence_hub.py sample_docs/neural_networks.pdf --budget 0.30 --eval


# With your own PDF from anywhere

python project_intelligence_hub.py "C:\Users\rajsr\Documents\my_paper.pdf"
```

CLI options:

Flag	Default	Description
pdf (positional)	required	Path to PDF file
--budget	0.50	Cost budget in USD

Flag	Default	Description
--retries	2	Max retry attempts if quality is low
--eval	off	Run evaluation harness
--no-cache	off	Disable semantic caching

4. What It Does

It runs a **7-agent pipeline** on your PDF:

Planner → [Summarizer | Fact Extractor | Quiz Generator | Gap Analyzer] (parallel)

→ Critic (quality gate) → Report Compiler

The output includes an executive summary, key facts, practice quiz, gap analysis, quality scores, and a cost breakdown.

5. First Run Note

The first run will download the sentence-transformers embedding model (~400MB) - this only happens once and runs locally (free, no API key needed).